

Predicting Pilot Intervention for Station-Keeping USVs: A Review

Jake Cushway
University of Victoria
jakecush1@uvic.ca

Rowan Hall
University of Victoria
rowanhall@uvic.ca

Abstract

This paper reviews the history and current state of human-monitored and autonomous station keeping for Uncrewed Surface Vehicles (USVs), focusing on whether the decision of *when human intervention is needed* can be automated. The Open Ocean Robotics (OOR) DataXplorer™ holds its position within 10 nm of a retired NOAA buoy station, with a pilot on 24/7 watch. Station-keeping control holds position but does not predict when a human is needed. Drift prediction and dynamic risk assessment give early warnings, but do not learn from past data when a human must act. We propose a probability-based solution that uses live telemetry to predict when a pilot is needed. Code and related documents are available at <https://github.com/jakecush1/autonomous-pilot-ML>.

1 Introduction

Station keeping is the task of holding a vessel inside a bounded region around a fixed point despite wind, waves and current. The DataXplorer, which replaces a retired weather buoy, need only stay within 10 nm of the buoy’s position; a shore-based pilot on 24/7 watch issues a course correction when needed, usually when wind pushes it toward the edge. Station keeping here has two layers: a *control* layer that decides *how* the vessel holds position, and a *supervision* layer that decides *when* a human must step in. **This work addresses the supervision layer: predicting from live telemetry whether a pilot will need to intervene within a future time window.**

1.1 History

The U.S. Coast Guard began holding station with crewed vessels to collect meteorological data in 1940 [2]. In the 1970s these ships were replaced by moored buoys, which hold station *passively*, on an anchor with no power or crew [2]. Offshore drilling, meanwhile, had automated *active* station keeping, using thrusters, with dynamic positioning in the 1960s [14].

USVs have been in development for over 100 years, but the ability to drive them autonomously is relatively new. Tesla demonstrated a radio-controlled boat in 1898, the Imperial German Navy used wire-guided explosive boats in the First World War, and by 1945 the U.S. Navy used remote-controlled boats for minesweeping and as targets, all within a short range of their operator [4]. These boats were all remotely controlled until MIT Sea Grant’s ARTEMIS (1993), among the first autonomous surface craft, and by the mid-2000s USVs had matured enough to have an impact in mission areas [7]. Long-endurance USVs such as the Wave Glider [6] followed, and USVs can now stand in for retired buoys, as the DataXplorer does for Station 46012. Using machine learning to emulate a human’s decisions when piloting marine vessels, however, is still a new concept.

Table 1: Related work by question and model type. DP: dynamic positioning; BN: Bayesian network; NN: neural network.

Question	Model-based	Data-driven
Hold station	Geofence; DP [14]; feedback [10, 11]	ML control [12]
Drift	Time to grounding [3]	Physics + NN [13]
When to act	Hazard ID [8]; dynamic BN [5]	Cars [9]; marine: this work

2 Related Work

Work related to supervised station keeping answers one of three questions (Table 1): *how* to hold a vessel on station (station-keeping control, §2.1); *where and how soon* wind and current will carry it (drift prediction, §2.2); and *when* a mission is at risk and a human should act (risk assessment and operator supervision, §2.3). Within each question, methods are either *model-based*, built from physics or expert judgement, or *data-driven*, learned from logged observations. Most marine work is model-based. One finding shapes our approach: distance from a boundary is a weak warning signal on its own, because it ignores where wind and current are pushing the vessel [3].

2.1 Station Keeping Control

The simplest practice is a trigger rule: the vessel may drift freely inside some radius and is corrected only when it nears the edge; this is known as *geofencing*. OOR’s pilots apply a similar practice at a larger scale, correcting the DataXplorer’s navigation as it nears its 10 nm boundary. *Feedback control* instead corrects continuously. Dynamic positioning adjusts thrust to hold large ships within metres of a set point [14]. Sarda et al. [11] applied the same idea to a 4 m USV, testing three controllers (proportional-derivative, back-stepping and sliding mode), with and without wind measurements as an extra input. USVs that cannot thrust sideways instead point their bow into the combined push of wind, waves and current [10]. Feedback controllers hold position with no human involved, but none predicts whether a correction will be needed later. Geofences are cheap and easy to understand, but they react to distance alone and ignore where the vessel is heading.

2.2 Drift Prediction

Drift prediction estimates where wind and current will carry a vessel that is not being steered, such as a ship that has lost propulsion. Dugan and Utne [3] use it to compute *time to grounding* (TTG), the expected time until such a ship drifts aground, and update it along the voyage as an early warning. Unlike earlier models, which assumed a fixed drift speed in the direction of the wind, theirs adds ocean current and wave forces from historical forecast data and treats drift direction as uncertain. If the hazard is the DataXplorer’s 10 nm boundary instead of shallow water, TTG becomes the time

Table 2: Representative work compared with our project. HIL: human in the loop.

Work	Inputs	Output	Learned	HIL
Sarda et al. [11] 4 m USV	Position, measured wind	Thruster commands	No	No
Dugan & Utne [3] Ship	Position, forecast wind, wave, current	Time to grounding	No	No
Song et al. [13] Drifting craft	GPS fixes, forecast wind, current	Drift track	Hybrid	No
Kristensen et al. [5] 5 m USV	Weather, battery, comms	P(mission failure)	No	Yes
This work <i>DataXplorer</i>	Position, distance to station, wind	P(pilot must act within Δt)	Yes (V2)	Yes

until the vessel leaves its station. TTG is purely physics-based, though, and was tested on one research-ship voyage with no real groundings. Song et al. [13] show the value of learning from data: a small neural network corrects a physics drift model each time the vessel reports its position, cutting mean track error from 5.75 km to 0.41 km. Both methods predict a path or a time; neither predicts whether a human will need to act.

2.3 Dynamic Risk Assessment & Supervision

Qualitative hazard identification lists what could go wrong before a trial but produces no numbers; its authors note that systems which drastically reduce human intervention may need real-time risk assessment [8]. Kristensen et al. [5] make risk assessment *dynamic*: a Bayesian network updated every two hours gives the probability of mission failure for a 5 m USV, which the authors suggest could tell a shore operator when to pay more attention. Its probabilities, however, come from experts and assumptions, and it was checked against a single mission. On the human side, Bogg and Birrell [1] found that a visual alert reduced unnecessary human intervention in an automated vehicle, and DeepTake [9] learns from vehicle and driver data to predict whether and how quickly a driver will take over. We found no marine work that learns when a human must intervene from logged telemetry.

2.4 Comparative Analysis

Table 2 and Figure 1 compare the representative work. Station-keeping control acts over metres and minutes, with no human and no look-ahead. Drift prediction warns before a boundary is crossed, but needs forecast wind and current fields we do not have. Dynamic risk assessment comes closest, estimating live mission risk for a supervised USV, but from expert-set probabilities. Our project sits where all three meet: a learned, telemetry-driven estimate of when a pilot must act to keep the vessel inside the 10 nm boundary.

3 Project Position

Our project works only at the supervision level; holding station remains the job of the vessel’s existing autopilot [11]. We build on the time-to-event warning of Dugan and Utne [3] and the probabilistic risk framing of Kristensen et al. [5], adopt the learned-correction idea of Song et al. [13], and take the distance-from-station geofence rule as the baseline our model must beat.

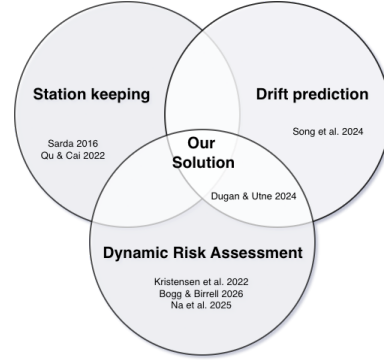


Figure 1: Previous papers relative to proposed project

- (1) **Data-driven prediction.** A model learns from logged telemetry (position, distance to station, wind speed and direction) the probability that the vessel will need a corrective command within a horizon Δt of about an hour.
- (2) **Reduced pilot labour.** This turns a continuous 24/7 watch (about 1,095 eight-hour shifts a year) into on-call, alert-driven work.
- (3) **Human integration.** A probability threshold sets when to alert the pilot, trading missed interventions against false alerts. The pilot makes every navigation decision.

References

- [1] Adam Bogg and Stewart Birrell. 2026. Do Not Adjust Your Set! How a Visual Alert Reduced Unnecessary Human Intervention in an Automated Vehicle. In *Proc. IHSI 2026*. 657–666. doi:10.54941/ahfe1007124
- [2] Robertson Dinsmore. 1996. Alpha, Bravo, Charlie.... *Oceanus* 39, 2 (1996). <https://www.who.edu/?p=21599>
- [3] Spencer August Dugan and Ingrid Bouwer Utne. 2024. Development of a Risk Indicator for Ship Drifting Groundings. In *Proc. PSAM17 & ASRAM2024*. <https://www.iapsam.org/PSAM17/program/Papers/PSAM17&ASRAM2024-1377.pdf>
- [4] H. R. Everett. 2015. *Unmanned Systems of World Wars I and II*. MIT Press.
- [5] Susanna Dybwad Kristensen, Yiliu Liu, and Ingrid Bouwer Utne. 2022. Dynamic Risk Analysis of Maritime Autonomous Surface Ships. In *Proc. PSAM 16*. <https://www.iapsam.org/PSAM16/papers/SU247-PSAM16.pdf>
- [6] Justin Manley and Scott Willcox. 2010. The Wave Glider: A Persistent Platform for Ocean Science. In *OCEANS’10 Sydney*. doi:10.1109/OCEANSSYD.2010.5603614
- [7] Justin E. Manley. 2008. Unmanned Surface Vehicles, 15 Years of Development. In *OCEANS 2008*. doi:10.1109/OCEANS.2008.5152052
- [8] Seong Na et al. 2025. Qualitative Risk Assessment Methodology for Maritime Autonomous Surface Ships: Cognitive Model-Based Functional Analysis and Hazard Identification. *J. Mar. Sci. Eng.* 13, 5 (2025), 970. doi:10.3390/jmse13050970
- [9] Erfan Pakdamanian et al. 2021. DeepTake: Prediction of Driver Takeover Behavior using Multimodal Data. In *Proc. CHI ’21*. doi:10.1145/3411764.3445563
- [10] Yang Qu and Lilong Cai. 2022. Nonlinear Station Keeping Control for Underactuated Unmanned Surface Vehicles to Resist Environmental Disturbances. *Ocean Eng.* 246 (2022), 110603. doi:10.1016/j.oceaneng.2022.110603
- [11] Edoardo I. Sarda et al. 2016. Station-Keeping Control of an Unmanned Surface Vehicle Exposed to Current and Wind Disturbances. *Ocean Eng.* 127 (2016), 305–324. doi:10.1016/j.oceaneng.2016.09.037
- [12] Armando J. Sinisterra et al. 2020. Nonlinear and Machine-Learning-Based Station-Keeping Control of an Unmanned Surface Vehicle. In *Proc. OMAE 2020*. doi:10.1115/OMAE2020-19276
- [13] Yuxuan Song et al. 2024. The Prediction and Dynamic Correction of Drifting Trajectory for Unmanned Maritime Equipment Based on Fully Connected Neural Network (FCNN) Embedding Model. *J. Mar. Sci. Eng.* 12, 12 (2024), 2262. doi:10.3390/jmse12122262
- [14] Asgeir J. Sørensen. 2011. A Survey of Dynamic Positioning Control Systems. *Annu. Rev. Control* 35, 1 (2011), 123–136. doi:10.1016/j.arcontrol.2011.03.008